COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 11 — Requirements Elicitation Techniques & Stakeholder Analysis

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Section 14: Classroom Activity — Plan an Elicitation Study

Context: CEMS students report missing registration deadlines and duplicate bookings. Event coordinators complain that manual capacity tracking takes hours and causes gate discrepancies.

1. Stakeholder Register

Stakeholder Group	Project Goals	Influence	Interest	Engagement Strategy
Undergraduate Students	Discover events easily; instant mobile registration confirmation.	Low	High	Online Surveys & Prototype Testing
Student Club Leads	Promote workshops; manage team registrations and rosters.	Medium	High	Semi-structured Interviews
Faculty Activity Heads	Approve budgets and verify venue safety compliance.	High	Medium	Formal One-on-One Briefings
College IT Infrastructure	Ensure secure authentication and zero campus network downtime.	High	High	Technical Architecture Review
Campus Security / Ushers	Rapidly verify valid student entry at venue gates.	Low	Medium	On-site Workflow Observation
Dean of Student Affairs	Ensure equitable access and collect accredited participation stats.	High	High	Bi-weekly Milestone Steering

2. Elicitation Techniques & Justification

- **Technique 1: Semi-Structured Interviews:** Targeted at Club Coordinators. Justification: Qualitative, open-ended discussions uncover workflow bottlenecks in manual spreadsheet reconciliations that structured forms fail to reveal.

- **Technique 2: Large-Scale Student Surveys:** Targeted at 500+ campus students. Justification: Fast, statistically representative quantitative data on preferred notification channels (WhatsApp vs Email) and mobile device usage patterns.
- **Technique 3: Direct Workplace Observation:** Shadowing gate volunteers during a live campus seminar. Justification: Reveals real-world pain points like weak Wi-Fi at auditorium gates and long ticket search times.

3. Elicitation Instruments (Interview Guide & Survey)

- **Interview Question 1:** "Walk me through the exact procedure you use when an event reaches capacity and a student requests urgent manual entry?"
- **Interview Question 2:** "How do you track attendance discrepancies between registered students and actual seated attendees?"
- **Interview Question 3:** "What specific data points do you need in the post-event summary report for faculty filing?"
- **Survey Question 1:** "How do you currently find out about upcoming campus events? [Notice Board / WhatsApp / Instagram / College Email]"
- **Survey Question 2:** "Have you ever experienced a duplicate registration error? [Never / Once / Multiple Times]"

4. Candidate Requirements Elicited

- **Candidate REQ-A:** The portal shall feature a live countdown timer showing remaining seats on the public event card.
- **Candidate REQ-B:** The system shall provide an offline-capable Progressive Web App mode for gate volunteer QR scanning.
- **Candidate REQ-C:** The system shall automatically generate participation certificates upon verified gate check-in.

Section 17: Homework — Elicitation Plan for a Centralized Library System

Requirements elicitation strategy for modernizing an automated University Library System.

- **1. Five Stakeholder Cohorts:** (a) Undergraduate Borrowers; (b) Research Scholars; (c) Circulation Counter Desk Staff; (d) Cataloging Librarians; (e) University Chief Information Officer.
- **2. Circulation Staff Interview Guide:** Six focused questions exploring peak checkout bottlenecks, barcode scanner read failures, fine dispute resolution, inter-library loan tracking, damaged book flagging, and shelf re-shelving reconciliation.
- **3. Student Survey Questions:** Four quantitative questions measuring digital catalog usage frequency, preferences for automated SMS due-date alerts, remote book reservation utility, and digital study room booking demand.
- **4. Observation Protocol:** Conduct a 2-hour observation at the central return desk during end-of-semester exam week to time transaction speeds and log manual register override frequency.
- **5. External System Interface Requirements:** (a) Interface with University Registrar Student Information System (SIS) to automatically sync enrollment status; (b) Interface with Campus Smart Card RFID payment switch for automated fine deduction.

- **6. Stakeholder Conflict Resolution Strategy:** Conflict: Research scholars demand a 90-day loan period for reference texts, while undergraduate students demand 7-day limits to maintain textbook circulation. Resolution: Establish an algorithmic tiered circulation rule allowing 30 days for general titles with an automated recall trigger if three or more reserve holds are queued.

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